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Dear Hiring Team at Tailscale,

I'm John Duncan, a senior software engineer who helped scale Quorum Analytics—the best-in-class civic tech and public affairs platform—from a bootstrapped startup to a global company serving over 2,000 clients, including over 50% of the Fortune 100. I thrive on solving ambiguous problems, balancing pragmatism while maintaining high standards. A former coworker once said that “[my] work really embod[ies] the idea that no problem is unsolvable, especially when you seek first to understand.” My natural curiosity drives me to deeply understand systems, leading to clean, efficient, and maintainable code that can be easily documented, maintained, extended, and deleted. I'm excited about the opportunity to bring my extensive full-stack expertise to Tailscale's marketing team.

At Quorum, I consistently delivered large-scale, cross-functional, and high-impact projects on time and under budget. As a technical lead, I drove growth by consistently solving the team's most complex engineering challenges.

- Led large-scale, cross-functional projects:** I recently launched PAC Hybrid, expanding Quorum into the \$15M+ PAC market and requiring constant cross-functional communication between engineers, product managers, management, QA engineers, professional services, marketing, and leadership teams to deliver a complex integration under a tight deadline. In addition to the technical research and implementation, this involved coordinating across teams, managing stakeholder expectations, and ensuring consistent communication throughout the development lifecycle, where I:
 - Discovered critical discrepancies in Federal Election Commission (FEC)-designated national party committee C-number ids officially listed across 50+ years of legal advisory opinions, FEC website data, and official fec-cms GitHub repository.
 - Developed end-to-end search, filtering, and data visualization features for four datasets, including backend sql queries, API endpoints, frontend UI functionality, and the technical design document (TDD) for the Dashboards integration.
 - Designed a service to instantly synchronize billions of rows between the Quorum and CisionPAC databases.
 - Reduced database query latency from 60 seconds to 20 milliseconds by optimizing query plan for partition and schema constraints, improving transaction dataset query performance by over 3000x and fixing a 10+ year old bug.
 - Coached engineers to optimize a database migration by replacing sequential queries with a single constant-time operation, reducing execution time from two weeks to under five seconds and improving backfill performance by over 240,000x.

Other projects of similar scope and complexity are listed in my full CV: <https://johnduncan.io/cv.pdf>.

- Thrived in fast-moving environments:** At Quorum, I consistently delivered projects on time and under budget in a high-growth startup environment, often working on multiple initiatives simultaneously. I'm accustomed to adapting quickly to changing requirements and priorities.
- Balanced strategic vision with hands-on implementation:** I've authored numerous technical design documents (TDDs), conducted user interviews with the design team, and worked closely with the product team on a sometimes daily basis to define product requirements documents (PRDs), all while working as a technical lead writing 5k+ commits, authoring 700+ pull requests, and performing 1000+ code reviews.
- Built and optimized full-stack infrastructure:** As technical lead for multiple flagship products, I developed products and features that balanced performance and user experience. I've worked extensively with Django, Python, PostgreSQL, SQL, JavaScript, and TypeScript, enabling me to understand the architectural considerations necessary to create beautiful, performant, accessible, scalable, and maintainable web experiences.
- Created reusable UI components:** I collaborated with our design team and founded the full-stack guild in order to write Quorum's first design system while working on the Dashboard visualization project, creating modular, reusable widgets and visualization components which maintain consistency and performance across the platform while improving developer efficiency.

A portion of my major work at Quorum is highlighted in my portfolio: <https://johnduncan.io/portfolio#quorum>.

I'm particularly drawn to Tailscale's mission of building a better internet and your commitment to creating a kind, thoughtful team culture. I believe my combination of technical expertise, leadership experience, ability to work effectively with cross-functional stakeholders, and expertise in leading projects from from conception to execution makes me an ideal candidate for this position.

I'm excited about the opportunity to apply my skills to help scale Tailscale's web presence and contribute to the company's continued growth. I would love to discuss how my background aligns with the position and how I can help drive Tailscale's web strategy forward.

Thank you for considering my application!

Sincerely,

